

AI Web Tester - API Migration Work Summary

Date: 22 May 2026 Project: AI Web Tester Location: ai-web-tester/backend/

Issue 1 - Anthropic API: Credit Balance Too Low

ERROR

The AI Web Tester uses Claude (Anthropic API) to drive browser actions. The backend returned HTTP 400 with the message: 'Your credit balance is too low to access the Anthropic API.' This blocked every test run immediately.

Root Cause:

- * Anthropic API key had zero remaining credits.

Options Considered:

- * Top up Anthropic credits (simplest, no code change)
- * Switch to Google Gemini free tier
- * Switch to Groq free tier
- * Use local Ollama model
- * Use AWS Bedrock / Google Vertex AI

Decision:

Switch to Google Gemini 2.0 Flash (free tier) - supports multimodal (images + text), minimal code change required.

Fix 1 - Migrated agent.py from Anthropic to Google Gemini

FIXED

Files Changed:

- * backend/agent.py
- * backend/.env (ANTHROPIC_API_KEY -> GEMINI_API_KEY)
- * backend/.env.example

Key Code Changes in agent.py:

```
# Before
import anthropic
self.client = anthropic.Anthropic(api_key=api_key)
self.model = 'claude-sonnet-4-6'

# After
import google.generativeai as genai
import PIL.Image
genai.configure(api_key=api_key)
self.model = genai.GenerativeModel(
    model_name='gemini-2.0-flash',
    system_instruction=SYSTEM_PROMPT,
    generation_config=genai.types.GenerationConfig(max_output_tokens=512),
)
```

Conversation Format Change:

```
# Before (Anthropic)
{'role': 'user',          'content': content_blocks} # list with image+text
```

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```
{'role': 'assistant', 'content': ai_text}
```

After (Gemini)

```
{'role': 'user', 'parts': [PIL.Image.open(path), user_text]}  
{'role': 'model', 'parts': [ai_text]}
```

Package Installed:

* google-generativeai (pip install google-generativeai)

Issue 2 - Gemini Free Tier Blocked in India (limit: 0)

ERROR

After switching to Gemini, every API call returned HTTP 429 with 'Quota exceeded' and limit: 0 for all free-tier metrics (GenerateRequestsPerDayPerProjectPerModel-FreeTier, etc.).

Tried creating a fresh API key from Google AI Studio - same result. The limit: 0 on every metric confirms the Gemini free tier is completely unavailable in the user's region (India). This is a Google infrastructure restriction, not a key or quota issue.

Decision:

Switch to Groq - completely free, no regional restrictions, 1,000 requests/day free tier.

Fix 2 - Migrated agent.py from Gemini to Groq (llama-3.3-70b-versatile)

FIXED

Files Changed:

- * backend/agent.py
- * backend/.env (GEMINI_API_KEY -> GROQ_API_KEY)
- * backend/.env.example

Key Code Changes in agent.py:

```
# Before (Gemini)  
import google.generativeai as genai, PIL.Image  
self.model = genai.GenerativeModel('gemini-2.0-flash', ...)  
response = self.model.generate_content(conversation)  
ai_text = response.text.strip()  
  
# After (Groq)  
from groq import Groq  
self.client = Groq(api_key=api_key)  
self.model = 'llama-3.3-70b-versatile'  
response = self.client.chat.completions.create(  
    model=self.model, max_tokens=512,  
    messages=[{'role': 'system', 'content': SYSTEM_PROMPT}] + conversation  
)  
ai_text = response.choices[0].message.content.strip()
```

Conversation Format (back to OpenAI-compatible):

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```
{'role': 'user',      'content': user_text}
{'role': 'assistant', 'content': ai_text}
```

Note: Groq does not support vision - screenshots are no longer sent to the AI. The agent still captures full page HTML which is sufficient for navigation and testing.

Package Installed:

- * groq (pip install groq)

Issue 3 - Groq Free Tier: Request Too Large (413)

ERROR

HTTP 413: 'Request too large for model llama-3.3-70b-versatile. Limit 12000 TPM, Requested 13417 tokens.'

Groq's free tier enforces a 12,000 tokens-per-minute limit. The conversation history grows with each step (up to 15 steps), and the page HTML (truncated at 4,000 chars) was too large when combined with accumulated message history.

Fix 3 - Reduced HTML size and capped conversation history

FIXED

Files Changed:

- * backend/browser.py - HTML truncation: 4,000 -> 2,000 chars
- * backend/agent.py - conversation history: full -> last 4 messages only

browser.py change:

```
# Before
return content[:4000]

# After
return content[:2000]
```

agent.py change:

```
# Before
messages=[{'role': 'system', 'content': SYSTEM_PROMPT}] + conversation

# After
trimmed = conversation[-4:] # keep last 4 messages (2 exchanges)
messages=[{'role': 'system', 'content': SYSTEM_PROMPT}] + trimmed
```

Token Budget After Fix (approx.):

- * System prompt: ~400 tokens
- * 4 history messages: ~1,400 tokens
- * Current user message: ~500 tokens
- * Total per request: ~2,300 tokens (well under 12,000 limit)

Final State - Working Configuration

Model:

- * Groq - llama-3.3-70b-versatile (free tier, no regional restriction)

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Environment Variable Required:

```
GROQ_API_KEY=gsk_... # in backend/.env
```

Get API Key:

* <https://console.groq.com> -> API Keys -> Create API Key (free signup)

Free Tier Limits (Groq):

- * 1,000 requests / day
- * 30 requests / minute
- * ~12,000 tokens / minute on free on-demand tier

Packages Installed in venv:

- * google-generativeai (installed but no longer used)
- * Pillow (installed but no longer used)
- * groq (active)

Change Log Summary

File	Change	Reason
agent.py	Anthropic -> Gemini	Anthropic credits exhausted

agent.py